

FORK INFOSYSTEMS

Karad, Maharashtra

OS INSTALLATION GUIDE

Windows 11 & Ubuntu Linux — Step-by-Step Setup (2025)

Covers	Windows 11 installation, Ubuntu Linux installation, bootable USB creation, BIOS/UEFI setup, post-install steps
Includes	System requirements, step-by-step wizards, partitioning guide, essential terminal commands & troubleshooting tips
Prepared For	Java Full Stack Developer Course Students
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PART 1 Windows 11 Installation

1.1 System Requirements

Component	Minimum	Recommended
Processor	1 GHz, 2+ cores (64-bit)	2 GHz multi-core
RAM	4 GB	8 GB or more
Storage	64 GB	256 GB SSD
Firmware	UEFI, Secure Boot capable	TPM 2.0 enabled
Display	720p, 9" diagonal	1080p or higher
Internet	Required for setup	Broadband

1.2 Download Windows 11 ISO

Download the official Windows 11 ISO from Microsoft's website. Use the Media Creation Tool for the simplest experience.

■ **Note:** Official link: microsoft.com/en-us/software-download/windows11

1.3 Create a Bootable USB Drive (Rufus)

- Insert a USB drive (8 GB minimum) into your computer
- Download and launch **Rufus** (rufus.ie)
- Select your USB drive under **Device**
- Click **SELECT** and choose the downloaded Windows 11 ISO
- Partition scheme: **GPT** • Target system: **UEFI (non CSM)**
- Click **START** and confirm to wipe the drive
- Wait for the process to complete (~10 minutes)

1.4 Boot from USB

Restart your PC and enter the BIOS/UEFI firmware. Common keys to enter BIOS:

Manufacturer	Key
Dell	F2 or F12
HP	F9 or Esc
Lenovo	F1 or F2
ASUS / MSI	Del or F2
Acer	F2 or Del

Set the USB drive as the first boot device, save changes, and reboot.

1.5 Windows 11 Setup Wizard

- **Language & Region:** Choose your language, time format, and keyboard layout, then click Next
- **Install Now:** Click the blue 'Install now' button on the welcome screen
- **Product Key:** Enter your license key or click 'I don't have a product key' to activate later
- **Edition:** Select the Windows 11 edition (Home or Pro) that matches your license
- **License Terms:** Accept the Microsoft Software License Terms and click Next
- **Installation Type:** Choose 'Custom: Install Windows only' for a clean install
- **Drive Selection:** Select the target partition or unallocated space. Delete old partitions if needed
- **Installing:** Windows copies files and installs features. The PC will restart several times
- **OOBE Setup:** Follow the Out-of-Box Experience wizard: region, keyboard, network, account
- **Done!** Windows 11 loads to the desktop. Install drivers via Windows Update

1.6 Post-Installation Checklist

- Run Windows Update (Settings > Windows Update > Check for updates)
- Install device drivers (Device Manager or manufacturer website)
- Enable Windows Defender / Security
- Create a System Restore point
- Install essential software (browser, office suite, etc.)
- Configure backup (Settings > System > Storage > Backup options)

PART 2 Ubuntu Linux Installation

2.1 System Requirements

Component	Minimum	Recommended
Processor	2 GHz dual-core	Multi-core 64-bit
RAM	4 GB	8 GB or more
Storage	25 GB	50 GB SSD
Display	1024 x 768	1920 x 1080
USB Port	USB 2.0	USB 3.0
Internet	Optional	Recommended

2.2 Download Ubuntu ISO

Download the latest Ubuntu LTS release from the official Ubuntu website. The LTS (Long Term Support) version is recommended for stability.

■ **Note:** Official link: ubuntu.com/download/desktop

2.3 Create a Bootable USB Drive

On Windows — using Rufus:

- Insert a USB drive (8 GB minimum)
- Open Rufus and select the USB device
- Click **SELECT** and choose the Ubuntu ISO
- Partition: **GPT or MBR** (match your firmware)
- Click **START** — write in ISO Image mode

On Linux / macOS — using terminal (dd):

```
# Find the USB device
lsblk

# Write ISO to USB (replace sdX with your device, e.g. sdb)
sudo dd if=ubuntu-24.04-desktop-amd64.iso \
of=/dev/sdX bs=4M status=progress oflag=sync
```

■ **Warning:** Double-check the target device with **lsblk** before running **dd**. Writing to the wrong device will erase its data.

2.4 Boot from USB

Restart your computer and press the boot-menu key (see Windows section for common keys). Select the USB drive. Choose **'Try or Install Ubuntu'** from the GRUB menu.

2.5 Ubuntu Setup Wizard

- **Try or Install:** Select 'Install Ubuntu' on the welcome screen
- **Language:** Choose your preferred language for the installation
- **Accessibility:** Configure accessibility options if needed
- **Keyboard:** Select your keyboard layout (auto-detect or manual)
- **Network:** Connect to Wi-Fi or Ethernet for downloading updates
- **Update & Software:** Choose 'Normal installation' and optionally install third-party software
- **Installation Type:** Select 'Erase disk and install Ubuntu' for a clean install, or choose 'Manual partitioning' for advanced setups
- **Disk Setup:** Confirm the partition changes and click 'Install Now'
- **Time Zone:** Select your location on the map or type a city name
- **User Account:** Enter your name, computer name, username, and a strong password
- **Installing:** Ubuntu copies and installs files (~15-25 minutes)
- **Restart:** Click 'Restart Now' and remove the USB drive when prompted

2.6 Post-Installation — Essential Terminal Commands

Open a terminal (Ctrl + Alt + T) and run the following:

```
# Update package lists and upgrade all packages
sudo apt update && sudo apt upgrade -y

# Install common utilities
sudo apt install -y curl wget git vim htop net-tools

# Install multimedia codecs
sudo apt install -y ubuntu-restricted-extras

# Check system info
neofetch # (install with: sudo apt install neofetch)
```

2.7 Manual Partitioning (Advanced)

Mount Point	Type	Size	Purpose
/boot/efi	FAT32	512 MB	EFI System Partition (UEFI only)
/	ext4	25 - 50 GB	Root filesystem
/home	ext4	Remaining	User data
swap	swap	= RAM size	Virtual memory

2.8 Post-Installation Checklist

- Run: **sudo apt update && sudo apt upgrade -y**
- Install GPU drivers (Software & Updates > Additional Drivers)
- Enable Ubuntu Firewall: **sudo ufw enable**
- Install Snap / Flatpak apps as needed
- Configure automatic updates (Software & Updates > Updates tab)
- Set up SSH if needed: **sudo apt install openssh-server**

PART 3 Troubleshooting & Tips

Q: Windows — TPM 2.0 / Secure Boot error

A: Enable TPM 2.0 and Secure Boot in BIOS/UEFI settings. On older boards, look for **PTT** (Intel) or **fTPM** (AMD) options.

Q: Windows — 'This PC can't run Windows 11'

A: Use the **PC Health Check** app to identify the specific issue. You may need to enable TPM, update BIOS, or upgrade hardware.

Q: Ubuntu — Black screen after boot

A: At GRUB, press **e**, find the 'quiet splash' line, replace with **nomodeset**, then press **F10** to boot. Install proprietary GPU drivers afterward.

Q: Ubuntu — Wi-Fi not detected

A: Run **sudo ubuntu-drivers autoinstall** to install proprietary drivers, or manually identify your chipset with **lspci | grep -i network**.

Q: Dual-boot GRUB not showing Windows

A: Boot into Ubuntu and run: **sudo os-prober && sudo update-grub**. Ensure os-prober is installed first.

Q: USB drive not booting

A: Verify the ISO integrity with an MD5/SHA256 checksum. Try a different USB port (preferably USB 2.0) and re-flash the drive.

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